

3.3 Methodology

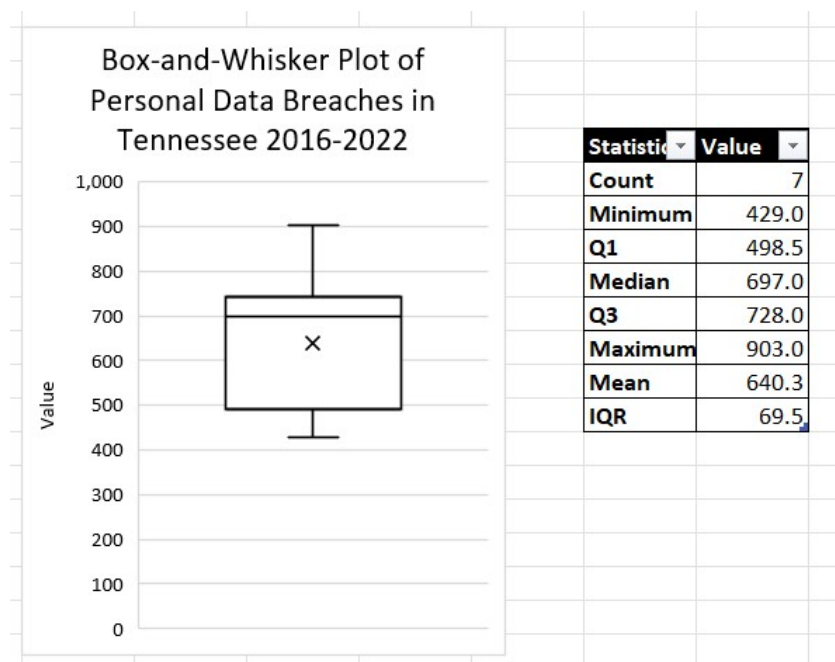


Fig. 3.3: Seven year summary of Personal Data Breaches in Tennessee reported to IC3.

We took cybercrime data from a seven year span of IC3 publications. The annual reports for the nation and for each state showed a count of victims per category of cybercrime. The US FBI defined each category for their report. We put that data into a matrix, organized by year and category of crime.

Each state we chose got its own matrix. We noticed that three states typically had high counts of crime; they were: California, Texas, and Florida. We noticed that US territories, like American Samoa, US Virgin Islands, and Puerto Rico, were reported also as if they were a state. These smaller areas with lower populations also showed low counts of cybercrime. We chose these high crime areas (e.g. California) and low crime areas (e.g. American Samoa) as our guides for choosing maximum and minimum limits. Our desired target areas were chosen by the target's home state; our initial runs included Chattanooga, Tennessee.

3.3.1 Using State Data to Forecast a Company's Data

Our main data sets are provided in the public interest; they're based on large units of people like states and nations. As system operators, we have a much smaller interest; we're interested in company sized elements. Instead of being concerned

with the cybercrime fate of over 300 million Americans or 9 million Tennesseans, we are probably more interested in what might happen to 100 stakeholders.

To smooth out the counts of cybercrime into rates, we adjusted these state victim counts per capita. We chose the US Census reports for those years to be the population values of the states. This would provide us with a per capita rate of cybercrime based on the chosen state which represented the chosen limit or target in the distribution.

Formally, we note this process below as shown in Figure 3.4. R represents our result. N_t represents our constant for target sample population and is used for multiplying the rate to scale it to match the population of the target. t represents our starting year for the sample from IC3 and coordinates the rows until it progresses to the final year chosen which is denoted by n . i represents our chosen category index of cybercrime. j represents our year index and is built with t until it progresses to the final year of the sample denoted by n . P_j represents the population of the target city at year j , drawn from matrix P .

$$R_j = N_t \sum_{j=t}^{t+6} \frac{X_{j,i}}{P_j}, \quad \text{where } X_{j,i} = \begin{cases} M_{j,i} & \text{if victim count} \\ V_{j,i} & \text{if victim impact} \\ S_{j,i} & \text{if suspect count} \\ D_{j,i} & \text{if suspect impact} \end{cases}$$

Fig. 3.4: Notation for the statistical calculation of the security event matrix adjusted for per-capita rates of incidence.

Five matrices inform this calculation. The first, P as shown in Figure 3.5, records the target city population for each year of the sample as a single column.

$$P = \text{Years} \left\{ \begin{array}{c} t \\ \vdots \\ j \\ \vdots \\ n \end{array} \right\} \left[\begin{array}{c} \text{Population} \\ \hline p_t \\ \vdots \\ p_j \\ \vdots \\ p_n \end{array} \right]$$

Fig. 3.5: Structure of the sample population matrix indexed by year.

The second matrix, M as shown in Figure 3.6, organises the raw cybercrime counts from IC3 by category and year, where each entry $a_{j,i}$ is a count that will be adjusted to a per-capita rate.

$$M = \text{Years} \left\{ \begin{array}{c} t \\ \vdots \\ j \\ \vdots \\ n \end{array} \right. \left[\begin{array}{c} \overbrace{\begin{array}{ccccccc} a_{t,1} & \cdots & a_{t,2} & \cdots & a_{t,i} & \cdots & a_{t,m} \\ \vdots & \ddots & \vdots & \ddots & \vdots & \ddots & \vdots \\ a_{j,1} & \cdots & a_{j,2} & \cdots & a_{j,i} & \cdots & a_{j,m} \\ \vdots & \ddots & \vdots & \ddots & \vdots & \ddots & \vdots \\ a_{n,1} & \cdots & a_{n,2} & \cdots & a_{n,i} & \cdots & a_{n,m} \end{array}}^{\text{Victim Counts by Categories}} \end{array} \right]$$

Fig. 3.6: Structure of the security event matrix for victim counts by cybercrime categories indexed by year.

The third matrix, V as shown in Figure 3.7, organises the victim impact in USD from IC3 by category and year, where each entry $a_{j,i}$ is a count that will be adjusted later to a per-capita rate.

$$V = \text{Years} \left\{ \begin{array}{c} t \\ \vdots \\ j \\ \vdots \\ n \end{array} \right. \left[\begin{array}{c} \overbrace{\begin{array}{ccccccc} a_{t,1} & \cdots & a_{t,2} & \cdots & a_{t,i} & \cdots & a_{t,m} \\ \vdots & \ddots & \vdots & \ddots & \vdots & \ddots & \vdots \\ a_{j,1} & \cdots & a_{j,2} & \cdots & a_{j,i} & \cdots & a_{j,m} \\ \vdots & \ddots & \vdots & \ddots & \vdots & \ddots & \vdots \\ a_{n,1} & \cdots & a_{n,2} & \cdots & a_{n,i} & \cdots & a_{n,m} \end{array}}^{\text{Victim Impacts by Categories}} \end{array} \right]$$

Fig. 3.7: Structure of the security event matrix for victim impacts, as in dollars of damage, by cybercrime categories indexed by year.

The fourth matrix, S as shown in Figure 3.8, organises the count of suspects from IC3 by category and year. Each entry $a_{j,i}$ helps us to understand how many suspects are operating within an area of concern. This is a count that will be adjusted later to a per-capita rate.

$$S = \text{Years} \left\{ \begin{array}{c} t \\ \vdots \\ j \\ \vdots \\ n \end{array} \right. \left[\begin{array}{c} \text{Suspect Counts by Categories} \\ \hline a_{t,1} \quad \cdots \quad a_{t,2} \quad \cdots \quad a_{t,i} \quad \cdots \quad a_{t,m} \\ \vdots \quad \ddots \quad \vdots \quad \ddots \quad \vdots \quad \ddots \quad \vdots \\ a_{j,1} \quad \cdots \quad a_{j,2} \quad \cdots \quad a_{j,i} \quad \cdots \quad a_{j,m} \\ \vdots \quad \ddots \quad \vdots \quad \ddots \quad \vdots \quad \ddots \quad \vdots \\ a_{n,1} \quad \cdots \quad a_{n,2} \quad \cdots \quad a_{n,i} \quad \cdots \quad a_{n,m} \end{array} \right]$$

Fig. 3.8: Structure of the security event matrix for suspect counts by cybercrime categories indexed by year. These counts consider the number of actors who engage victims in a given area.

The fifth matrix, D as shown in Figure 3.9, organises the damages attributed by suspects in USD from IC3 by category and year, where each entry $a_{j,i}$ is a count that will be adjusted later to a per-capita rate. This helps us to understand rates of effective impact, per suspect, in a given area.

$$D = \text{Years} \left\{ \begin{array}{c} t \\ \vdots \\ j \\ \vdots \\ n \end{array} \right. \left[\begin{array}{c} \text{Suspect Damage by Categories} \\ \hline a_{t,1} \quad \cdots \quad a_{t,2} \quad \cdots \quad a_{t,i} \quad \cdots \quad a_{t,m} \\ \vdots \quad \ddots \quad \vdots \quad \ddots \quad \vdots \quad \ddots \quad \vdots \\ a_{j,1} \quad \cdots \quad a_{j,2} \quad \cdots \quad a_{j,i} \quad \cdots \quad a_{j,m} \\ \vdots \quad \ddots \quad \vdots \quad \ddots \quad \vdots \quad \ddots \quad \vdots \\ a_{n,1} \quad \cdots \quad a_{n,2} \quad \cdots \quad a_{n,i} \quad \cdots \quad a_{n,m} \end{array} \right]$$

Fig. 3.9: Structure of the security event matrix for suspect damage, in dollars by perpetrators, by cybercrime categories indexed by year. This tally helps us understand how damaging each actor is to victims.

We apply these matrices to the formula shown in Figure 3.4. The entries $a_{j,i}$ are the raw counts taken from the IC3 state-level reports for the chosen source state. Dividing each entry by P_j converts it to a per-capita rate; multiplying by N_t then scales that rate to match the population of the target city. The population values in P provide the year-by-year context for N_t and confirm the representativeness of the scaling. Summing the selected column i of the per-capita-adjusted M across all years yields the final result R .

3.3.2 Methodology for Monte Carlo Simulations in Excel

We chose IC3 crime data from American Samoa to act as our 10% confidence limit. We chose data from California to act as our 90% confidence limit. We set those limits by looking at seven year spans of crime data and taking the median. This means we would be likely to have observations above and below the limits, but that the limits themselves would have a basis in observation. We input and processed the data as follows:

- We imported the data from IC3 crime tables
- We calculated counts and costs for:
 - Chattanooga
 - American Samoa
 - California.
- We imported a normal distribution table from the Monte Carlo Toolkit.
- We set mean summary counts from American Samoa as our 10% interval.
- We set mean summary counts from California as our 90% interval.
- We accepted the Monte Carlo percentages and exceedence curves.

There were times when it was obvious that a normal distribution would not fit. When there was a choice between a count of 0 and 1 events, we ran a supplementary binary distribution. When there was a choice of less than 8 events, we ran a supplementary discrete distribution. We needed to change the distribution type when modeling the count or likelihood of an event; we did not need to change from a normal distribution when modeling impact because the range of currency was not overly narrow.

3.3.3 Population Changes During Sampled Years

Table 3.1: Changes to Population in Key Areas

Year	American Samoa Population	References	City of Chattanooga Population	References	State of Tennessee Population	References	State of California Population	References
2016	52,245	[8]	177,746	[9]	6,646,010	[10]	39,149,186	[11]
2017	51,586	[12]	179,530	[13]	6,708,799	[14]	39,337,785	[15]
2018	50,908	[16]	181,918	[17]	6,771,631	[18]	39,437,463	[19]
2019	50,209	[20]	182,799	[21]	6,829,174	[22]	39,437,610	[23]
2020	49,751	[24]	181,560	[25]	6,925,619	[26]	39,538,223	[27]
2021	49,225	[28]	181,163	[29]	6,968,351	[30]	39,152,927	[31]
2022	48,342	[32]	184,086	[33]	7,051,339	[34]	39,125,347	[35]

City population based on the US Census reported population for the City of Chattanooga itself. Surrounding satellite cities and unincorporated areas within the Hamilton County, Tennessee, were not included. Population for the State of Tennessee provided perspective on calculated rates in later tables. Population for American Samoa and the State of California were included for perspective on maximum and minimum counts in later tables.